

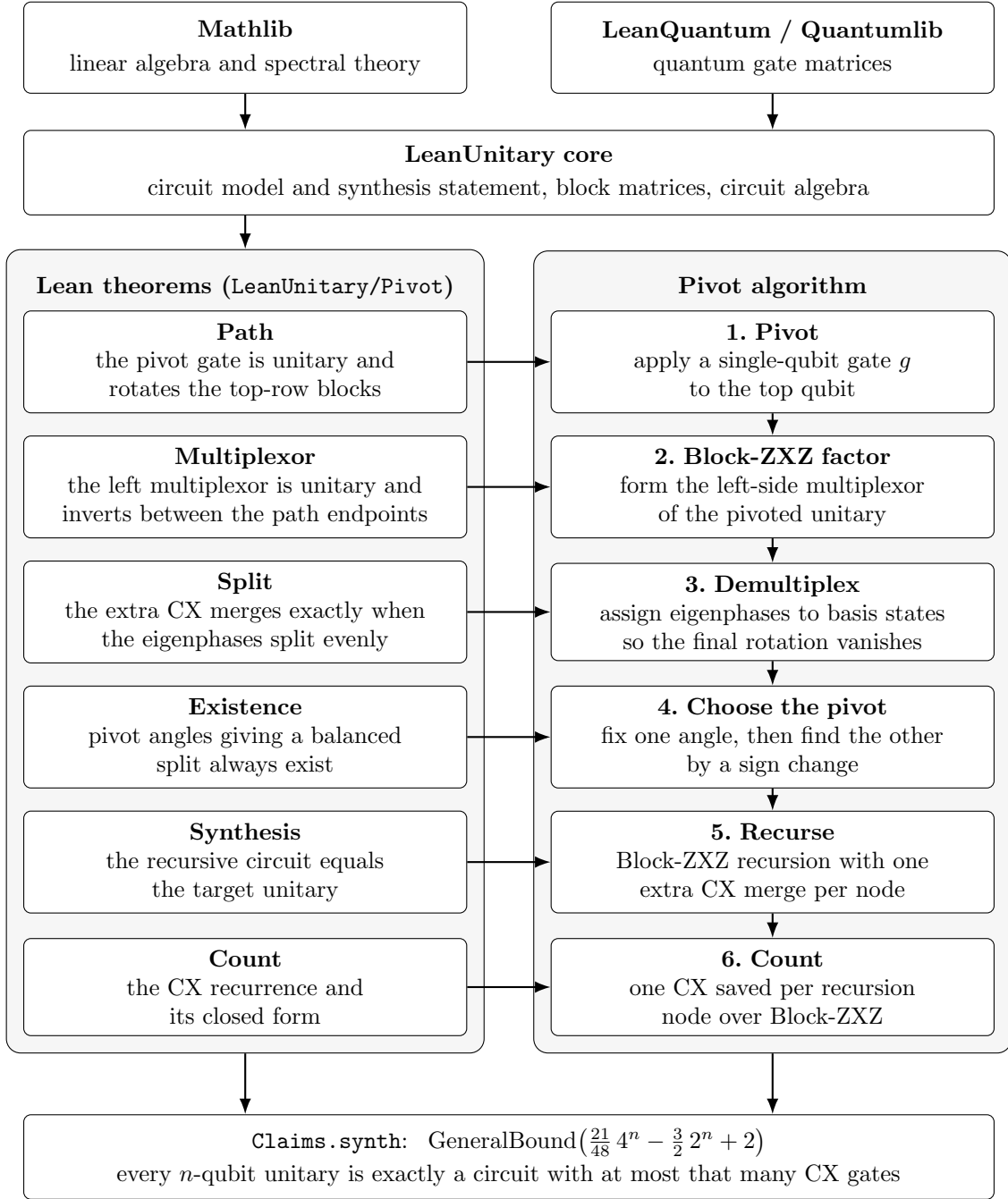


Figure 1: Lean formalization of our proposed pivot decomposition. Each theorem module on the left verifies the step of the pivot algorithm on its right using Mathlib’s linear algebra and the circuit model of LeanUnitary.